

Darren Cornelius Suwandi

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RESEARCH INTERESTS

Adversarial Machine Learning, Explainable AI (XAI), Behavioral Threat Detection, LLM Engineering, AI-Driven Full-Stack Systems.

EDUCATION

Xiamen University Malaysia

Sep. 2025 – Expected Aug. 2028

Bachelor of Digital Media and Technology | CGPA: 3.86/4.00 | Ranked #2 in Major

Selangor, Malaysia

- **Core Coursework:** Data Structure, Algorithm Analysis and Design, Software Architecture and Design Patterns, Operating Systems, Database, Principle of Artificial Intelligence, Big Data Analytics, Information Security, Software Project Management

RESEARCH EXPERIENCE

Xiamen University Malaysia

May 2026 – Present

Undergraduate Researcher (Supervised by Ts. Dr. Teh Jia Yew and Dr. Goh Sim Kuan)

Selangor, Malaysia

- Conduct research in adversarial machine learning and Explainable AI (XAI) to enhance the robustness of behavioral threat detection systems.
- Develop high-performance classification pipelines utilizing tree-based ensembles (**XGBoost**) and evaluate model resilience against feature-space evasion techniques.
- Design strict anti-leakage cross-validation protocols and optimize inference architectures for simulated edge deployment environments.

ENGINEERING PROJECTS

WhatIf Forecaster | *Next.js, FastAPI, FLAML, SHAP, Scikit-learn*

Apr. 2026 – Present

- Engineered a decoupled AutoML SaaS platform utilizing **FastAPI** and **FLAML** to automate model selection (e.g., Random Forest) with real-time training logs streamed via Server-Sent Events (SSE).
- Developed a highly reactive **Next.js** and Zustand dashboard featuring interactive forecasting simulations and **SHAP** integrations for local feature influence analysis.

Resume Forge | *Next.js, FastAPI, Gemini API, Supabase, Redis*

Feb. 2026 – Aug. 2026

- Architected a dual-phase ATS scoring engine and fault-tolerant Gemini API integration layer, blending deterministic Python extraction with strict LLM auditing for recruiter-aligned feedback.
- Developed an agentic editing suite featuring an SSE streaming chatbot and skill-gap analyzer, leveraging Zustand state management to instantly apply structured Before/After rewrite blocks.
- Implemented a comprehensive tiered credit system utilizing Redis for guest session tracking, Stripe for payment processing, and Supabase JWTs for secure API endpoint authentication.

Omni-QC SMT Quality Control System | *React, XGBoost, SHAP, IPC-2591 CFX*

Mar. 2026 – May 2026

- Directed a 7-person engineering team as Project Lead and Frontend Architect, managing a 3-person interdisciplinary advisory board to deliver a predictive defect-routing SaaS, reducing total AOI inspection workloads by 30–50% via real-time IPC-2591 CFX machine data evaluation.
- Engineered a high-performance **React** and **TypeScript** dashboard that bridges complex **XGBoost** and **SHAP** anomaly detection models with transparent, human-in-the-loop prescriptive recommendations for factory technicians.

KerjaCerdas Job-Matching Platform | *FastAPI, LangGraph, React, pgvector, Gemini API* Apr. 2026 – Jun. 2026

- As Product/UX Lead, architected a **LangGraph** ReAct Supervisor Swarm routing parallel worker agents for resume parsing, skill-gap analysis, and career advisory, alongside an E-KYC verification pipeline.
- Built a hybrid ranking engine (cosine, skill, region, salary fit) on **PostgreSQL (pgvector)** with Gemini embeddings, exposing 5-component explainability scores through a reactive **React/Zustand** dashboard.

AskA Maintenance Automation | *n8n, LLM, MySQL, Google Workspace APIs*

Mar. 2026 – Apr. 2026

- Collaborated in a 5-person team with Data Science and Electrical and Electronics Engineering (EEE) students, commissioned by Angie Ng (MQA - IT Services), to automate XMUM's maintenance infrastructure via **n8n**.
- Engineered intelligent email routing and automated worker assignment via Google Calendar while deploying an LLM conversational agent to triage requests and optimize workflow efficiency.

Orion Reimbursement System | *FastAPI, LangGraph, React, TypeScript, LLM*

Apr. 2026

- Led the frontend architecture within a 4-person team, delivering a role-based **React** and **TypeScript** dashboard that optimizes the user experience for end-to-end expense tracking and financial approvals.
- Integrated the UI with a **FastAPI** backend powered by a 6-agent **LangGraph** workflow, successfully eliminating manual triage bottlenecks through automated SaaS deduplication and deterministic policy checks.

Startup EMP | *FastAPI, LangGraph, Gemini API, Firestore, Cloud Run*

May 2026

- Built a deterministic, human-in-the-loop semantic engine within a 4-person team to safely evaluate mentor-founder pairings, mitigating LLM hallucinations via strict **Pydantic** schema enforcement and structured validation layers.
- Architected a serverless **FastAPI** application on **Google Cloud Run** and **Firestore**, integrating a **LangGraph**-orchestrated Gemini pipeline to autonomously parse multimodal pitch decks into structured JSON dossiers.

Study Buddy | *Flutter, Python, FastAPI, n8n*

Feb. 2026 – Mar. 2026

- Developed a cross-platform learning assistant integrating Gemini, GPT, and Claude APIs to support complex, multi-model conversational capabilities.
- Engineered automated data processing pipelines for robust PDF document parsing and dynamic YouTube video summarization.

Blackjack Q-Trainer | *Python, Reinforcement Learning, Pygame, NumPy*

Jan. 2026 – Apr. 2026

- Developed a Reinforcement Learning simulation, training a **Q-Learning** agent over 200,000 episodes featuring intelligent dynamic bankroll sizing via the Kelly Criterion and Hi-Lo card counting.
- Built a phase-guarded **Pygame** UI to visualize decision-making, integrating **NumPy** and **Matplotlib** to generate data-driven policy heatmaps from persistent Q-tables.

HONORS & AWARDS

School of Computing and Data Science Dean's List Award, Xiamen University Malaysia 2024, 2025, 2026

Full Tuition Scholarship, Xiamen University Malaysia 2024 – Present

Silver Award, CICSIC SEA (Omni-QC, Undergraduate Division) 2026

Finalist, Digdaya (KerjaCerdas) 2026

EXTRACURRICULARS & COMPETITIONS

Active Hackathon Competitor

2026

- Competed in SEA-CICSIC 2026, Digdaya 2026, UM Hackathon 2026, and MyHack Google Hackathon 2026 engineering rapid-viability full-stack AI prototypes under strict 24-to-48-hour constraints.
- Engineered rapid-viability full-stack AI prototypes under strict 24-to-48-hour competitive constraints.

Artificial Intelligence Club | *Member*

Sep. 2025 – Present

- Participated in technical discourse and collaborative initiatives focused on Git workflows, n8n automation, machine learning, and deep learning.

Coding Club | *Member*

Apr. 2026 – Present

- Engaged in practical engineering workshops focusing on CI/CD pipelines and full-stack web development.

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, JavaScript, SQL (MySQL)

Tools & DevOps: n8n, Unity, Blender, Android Studio, Wireshark, Weka

Frameworks: Next.js, FastAPI, PyTorch, Scikit-Learn, Optuna, SHAP, SMOTE, Pandas, Pydantic